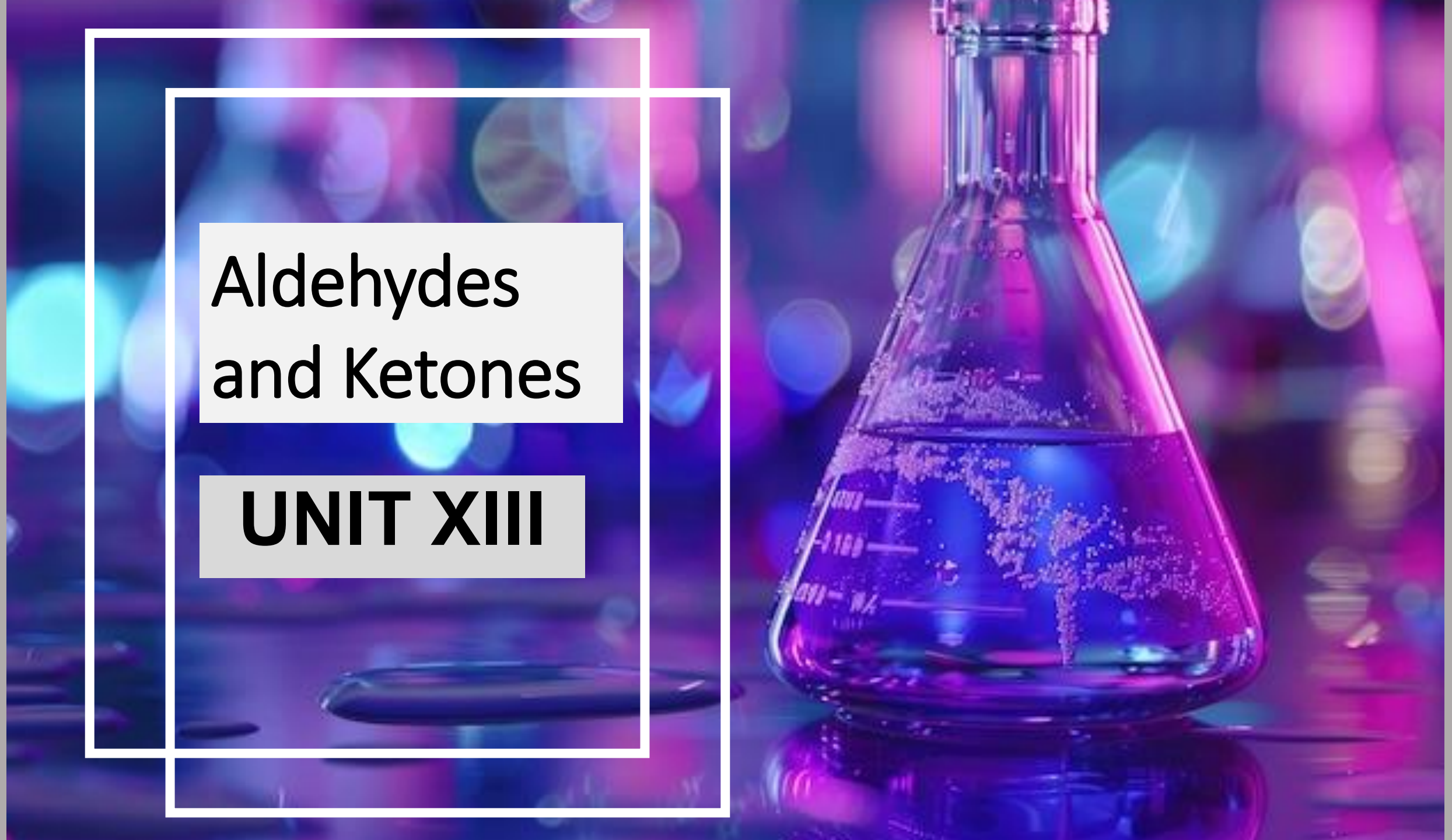


Aldehydes and Ketones

UNIT XIII

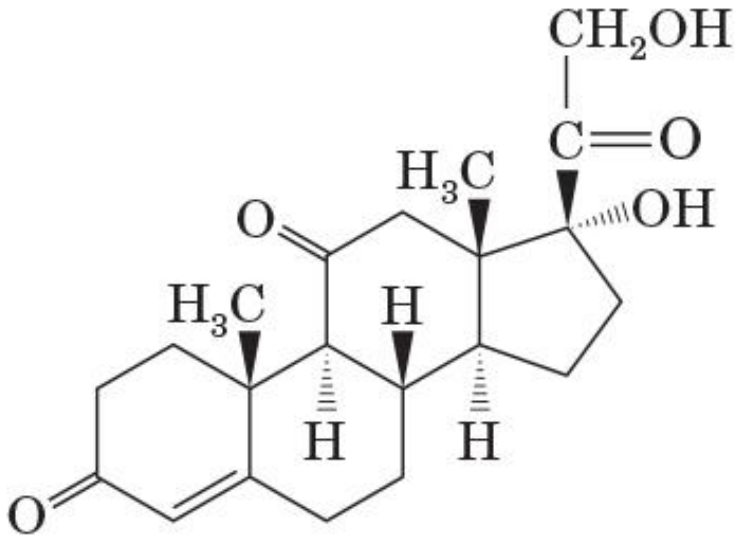
Learning Outcomes

At the end of each topic, the students should be able to:

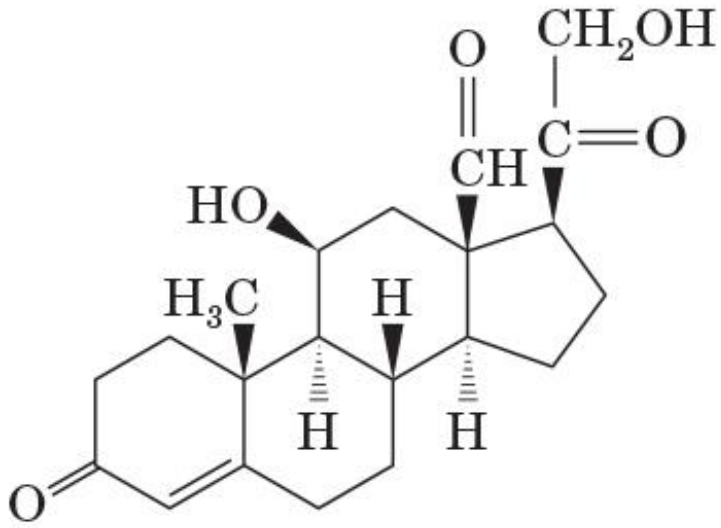
- ❑ Identify and classify organic compounds of aldehydes and ketones based on their structures and characteristic functional groups.
- ❑ Name and write the structural formulas of aldehydes and ketones using appropriate IUPAC nomenclature rules.
- ❑ Explain the physical properties of aldehydes and ketones.
- ❑ Describe the characteristic chemical reactions of aldehydes and ketones based on their functional groups.

EXERCISE

Here are structural formulas of two steroid hormones. Name the functional groups present in each.



Cortisone



Aldosterone

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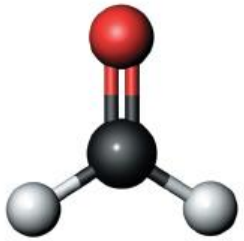
Isomers

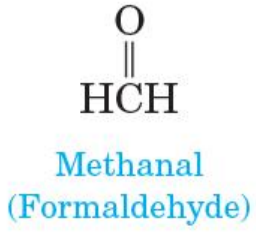
The functional group of an **aldehyde** is a carbonyl group bonded to

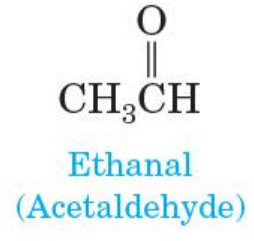
a hydrogen atom.

- In methanal (formaldehyde), the simplest aldehyde, the carbonyl group is bonded to two hydrogens.
- In other aldehydes, it is bonded to one hydrogen and one carbon group.

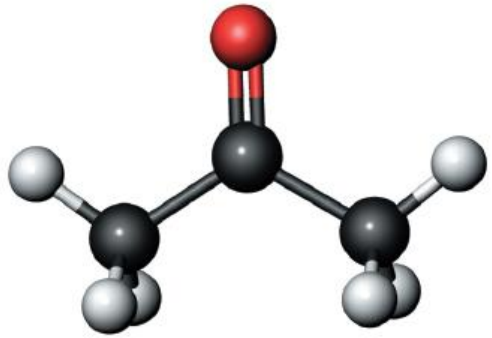
The functional group of a **ketone** is a carbonyl group bonded to two carbon groups.











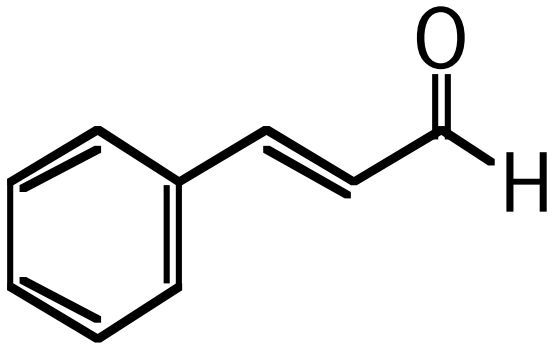
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Nomenclature

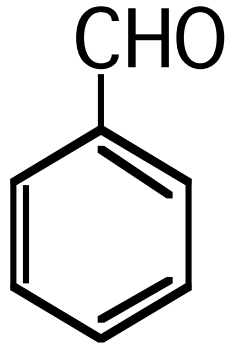
IUPAC names for aldehydes

- To name an aldehyde, change the suffix **-e** of the parent alkane to **-al**.
- Because the carbonyl group of an aldehyde can only be at the end of a parent chain and numbering must start with it as carbon-1, there is no need to use a number to locate the aldehyde group.
- For **unsaturated aldehydes**, indicate the presence of a carbon–carbon double bond by changing the ending of the parent alkane from **-ane** to -**enal**. Numbering the carbon chain begins with the aldehyde carbonyl carbon. Show the location of the carbon-carbon double bond by the number of its first carbon.

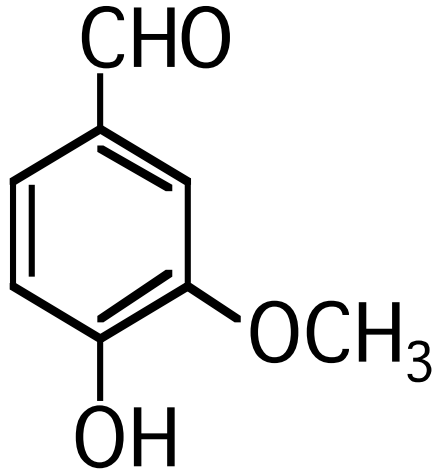
- The IUPAC system retains common names for some aldehydes, including these three.



***trans*-3-Phenyl-2-propenal (Cinnamaldehyde; in oil of cinnamon)**



Benzaldehyde (in almonds)

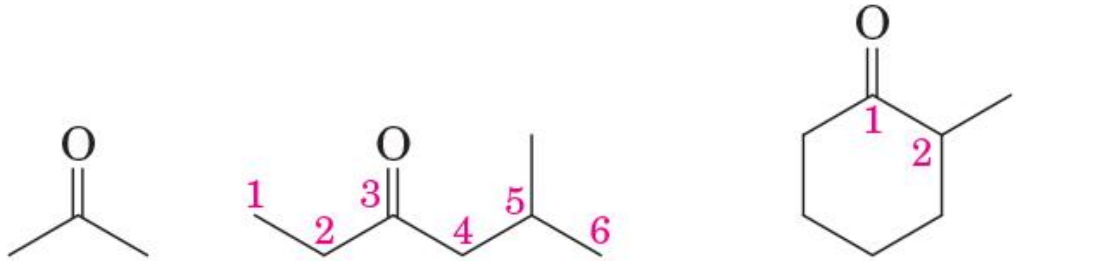


Vanillin (from vanilla beans)

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IUPAC names for ketones

- The parent alkane is the longest chain that contains the carbonyl group.
- Indicate the presence of the carbonyl group by changing the **-ane** of the parent alkane **-one**.
- Number the parent chain from the direction that gives the carbonyl carbon the smaller number.
- The IUPAC retains the common name acetone for 2-propanone.

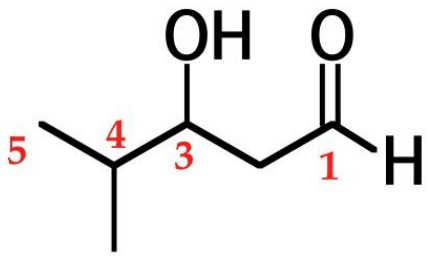


Acetone 5-Methyl-3-hexanone 2-Methylcyclohexanone

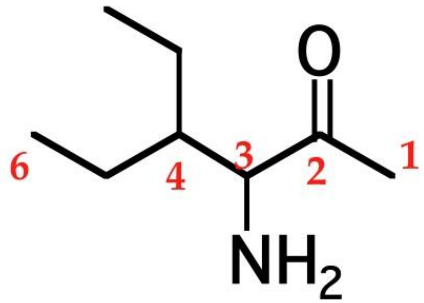
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To name an aldehyde or ketone that also contains an –OH (hydroxyl) or –NH_2 (amino) group:

- Number the parent chain to give the carbonyl carbon the lower number.
- Indicate an –OH substituent by **hydroxy-**, and an –NH_2 substituent by **amino-**.
- Hydroxyl and amino substituents are numbered and alphabetized along with other substituents.



3-Hydroxy-4-methylpentanal



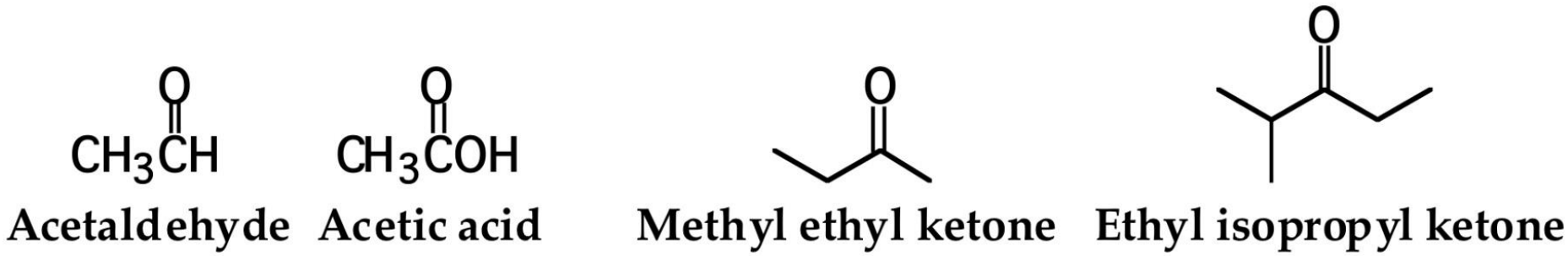
3-Amino-4-ethyl-2-hexanone

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Common names

The common name for an aldehyde is derived from the common name of the corresponding carboxylic acid.

- Drop the word "**acid**" and change the suffix **-ic** or **-oic** to **-aldehyde**.
  - Name each alkyl or aryl group bonded to the carbonyl carbon as a separate word, followed by the word "**ketone**." Alkyl or aryl groups are generally listed in order of increasing molecular weight.

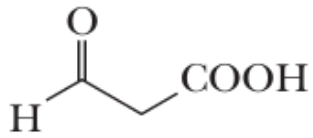
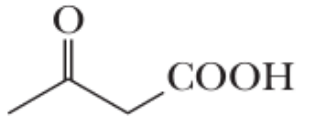
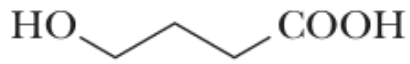
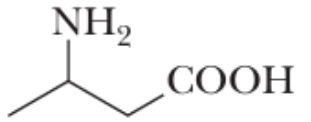
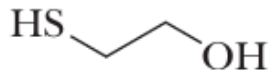


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Order of precedence of functions

Table 16.1 Increasing Order of Precedence of Six Functional Groups

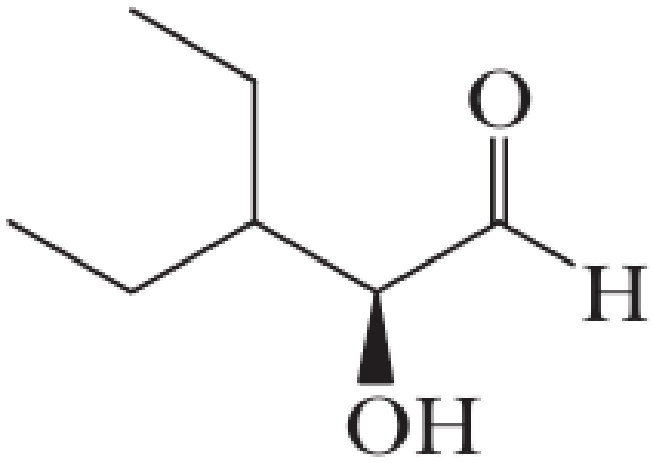
Increasing precedence

Functional Group	Suffix If Higher Priority	Prefix If Lower Priority	Example When the Functional Group Has Lower Priority
Carboxyl	-oic acid	—	
Aldehyde	-al	oxo-	3-Oxopropanoic acid 
Ketone	-one	oxo-	3-Oxobutanoic acid 
Alcohol	-ol	hydroxy-	4-Hydroxybutanoic acid 
Amino	-amine	amino-	3-Aminobutanoic acid 
Sulfhydryl	-thiol	mercapto-	2-Mercaptoethanol 

EXERCISE

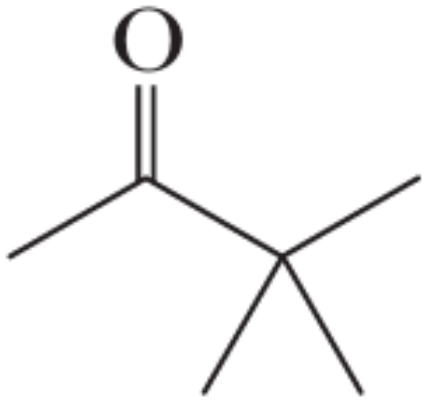
Give the IUPAC names for each compound.

a.



3-ethyl-2-hydroxypentanal

b.



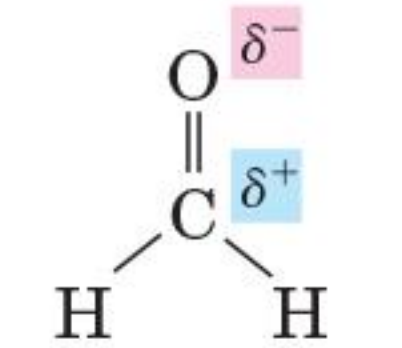
3,3-Dimethyl-2-butanone

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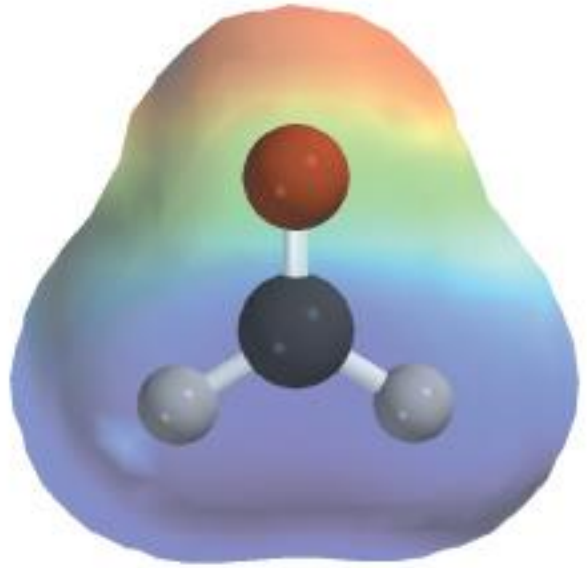
Physical Properties

A C=O bond is polar, with oxygen bearing a partial negative charge and carbon bearing a partial positive charge.

- Therefore, aldehydes and ketones are polar molecules.
- Figure 16.1 The polarity of a carbonyl group.



Polarity of a carbonyl group



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- In liquid aldehydes and ketones, there are weak intermolecular attractions between the partial positive charge on the carbonyl carbon of one molecule and the partial negative charge on the carbonyl oxygen of another molecule.
- No hydrogen bonding is possible between aldehyde or ketone molecules because they have no polar O–H or N–H bonds to participate in hydrogen bonding.
- Aldehydes and ketones have lower boiling points than alcohols and carboxylic acids, compounds in which there is hydrogen bonding between molecules. See the table on the next screen.

Table 16.1 Boiling Points for Six Compounds of Comparable Molecular Weight.

Name	Structural Formula	Molecular Weight (amu)	Boiling Point (°C)
diethyl ether	$CH_3CH_2OCH_2CH_3$	74	34
pentane	$CH_3CH_2CH_2CH_2CH_3$	72	36
butanal	$CH_3CH_2CH_2CHO$	72	76
2-butanone	$CH_3CH_2COCH_3$	72	80
1-butanol	$CH_3CH_2CH_2CH_2OH$	74	117
propanoic acid	CH_3CH_2COOH	74	141

- Formaldehyde, acetaldehyde, and acetone are infinitely soluble in water.
- Aldehydes and ketones become less soluble in water as the hydrocarbon portion of the molecule increases in size.

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EXERCISE

Comparison of the boiling points of aldehydes and ketones with those of other compounds of similar molecular mass shows that they are __.

a. Lower than those of alcohols and alkanes.

b. Higher than those of alcohols and alkanes.

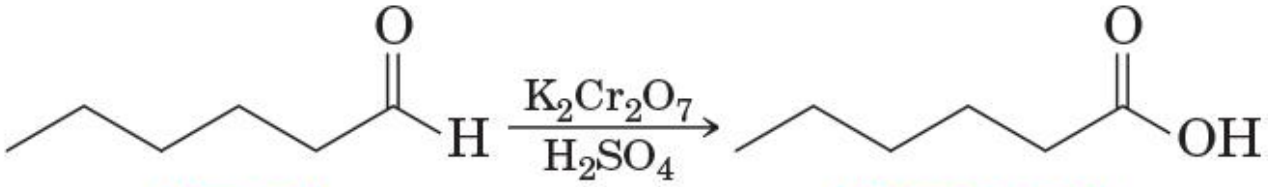
c. Higher than those of alcohols but lower than those of alkanes.

d. Higher than those of alkanes but lower than those of alcohols.

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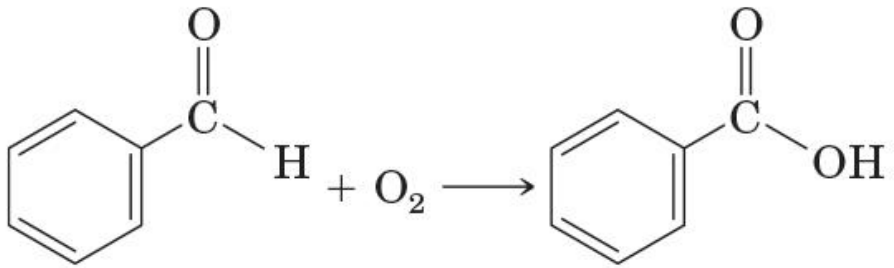
Oxidation

- Aldehydes are oxidized to carboxylic acids by a variety of oxidizing agents, including potassium dichromate.



Hexanal → Hexanoic acid

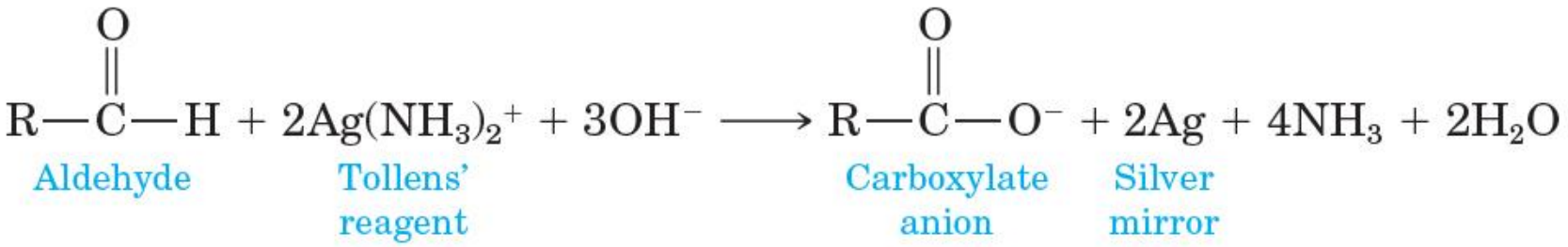
- Liquid aldehydes are so sensitive to oxidation by O_2 in the air that they must be protected from contact with air during storage.



Benzaldehyde → Benzoic acid

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- Ketones resist oxidation by most oxidizing agents, including potassium dichromate and molecular oxygen.
- Tollens' reagent is specific for the oxidation of aldehydes. If done properly, silver deposits on the walls of the container as a silver mirror.



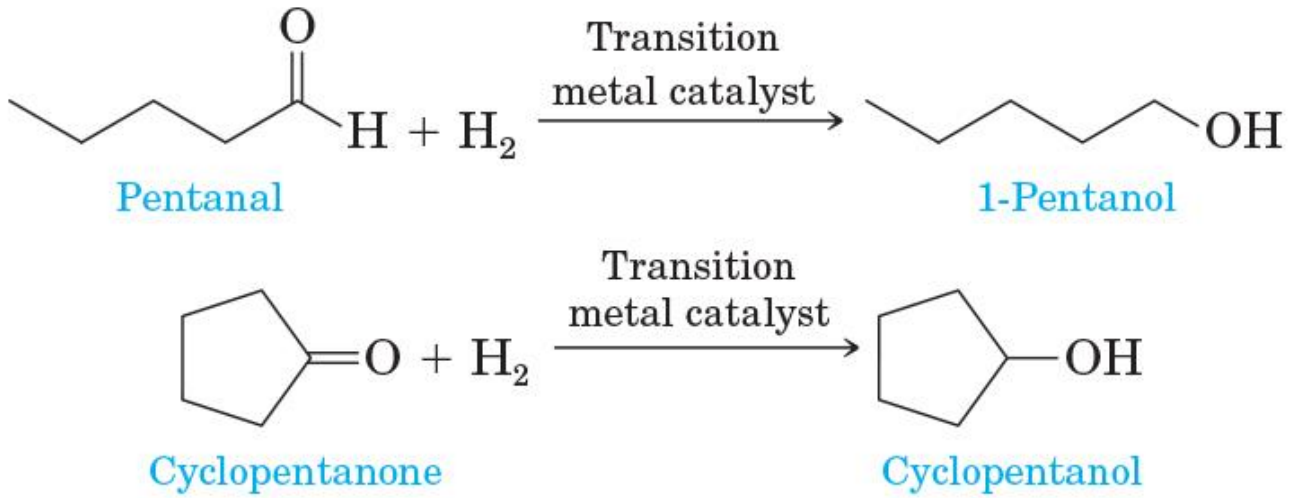


A silvered mirror has been deposited in the inside of this flask by the reaction of an aldehyde with Tollens' reagent.

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Reduction

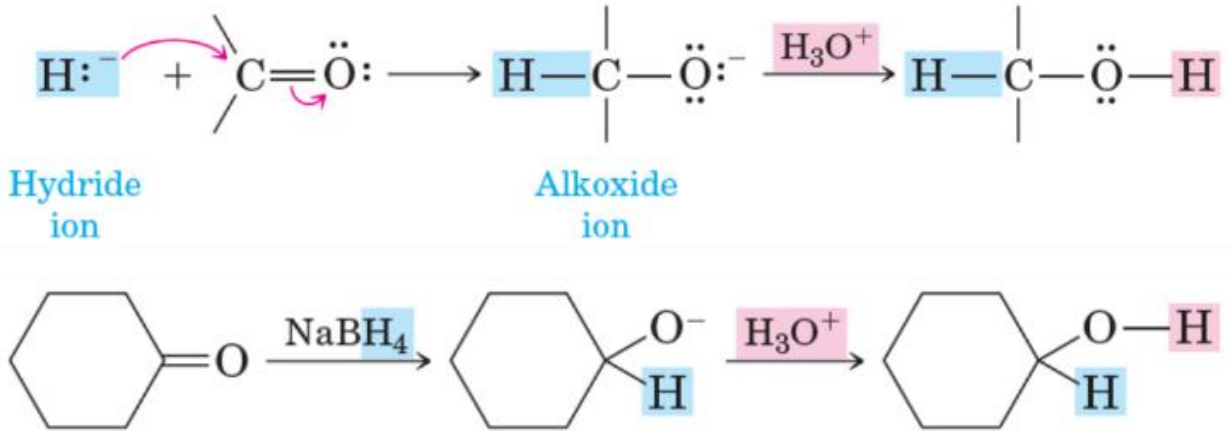
- The carbonyl group of an aldehyde or ketone is reduced to a –C(H)OH group by hydrogen in the presence of a transition-metal catalyst.
  - Reduction of an aldehyde gives a primary alcohol.
  - Reduction a ketone gives a secondary alcohol.



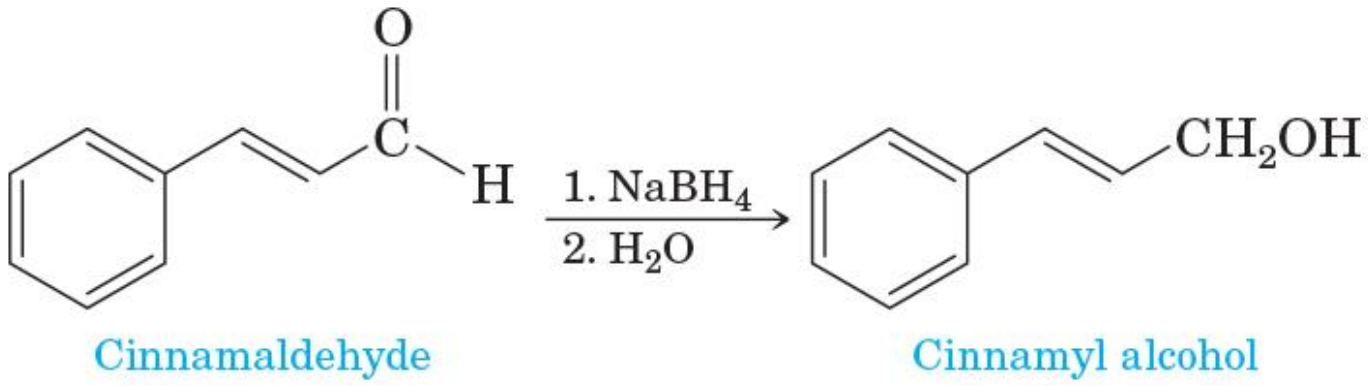
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The most common laboratory reagent for the reduction of an aldehyde or ketone is sodium borohydride, **$NaBH_4$**.

- This reagent contains hydrogen in the form of hydride ion, **$H:^-$**.
- In a hydride ion, hydrogen has two valence electrons and bears a negative charge.
- In a reduction by sodium borohydride, hydride ion adds to the partially positive carbonyl carbon which leaves a negative charge on the carbonyl oxygen.
- Reaction of this intermediate with aqueous acid gives the alcohol.

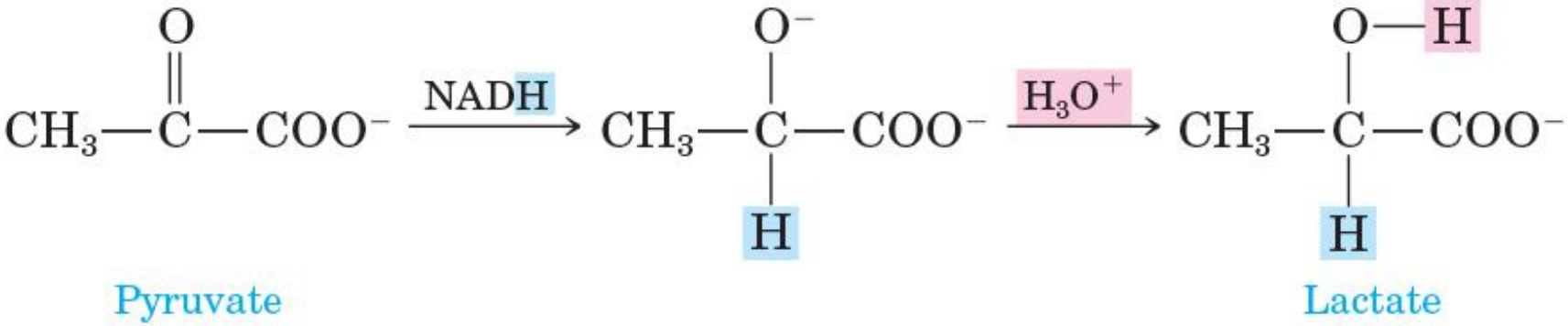


- Reduction by $NaBH_4$ does not affect a carbon-carbon double bond or an aromatic ring.



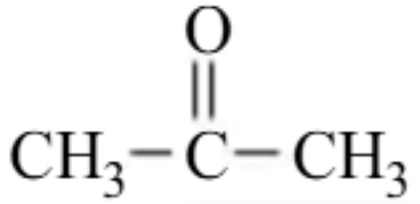
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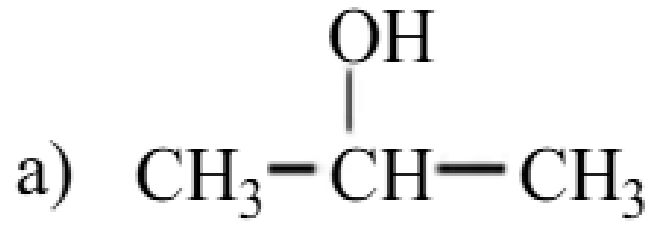
- In biological systems, the agent for the reduction of aldehydes and ketones is the reduced form of nicotinamide adenine dinucleotide, abbreviated NADH
  - This reducing agent, like $NaBH_4$, delivers a hydride ion to the carbonyl carbon of the aldehyde or ketone.
  - Reduction of pyruvate, the end product of glycolysis, by NADH gives lactate.

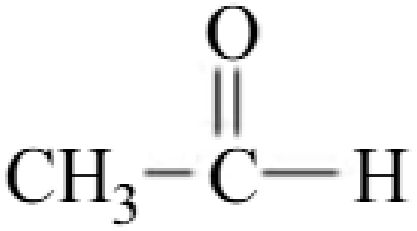


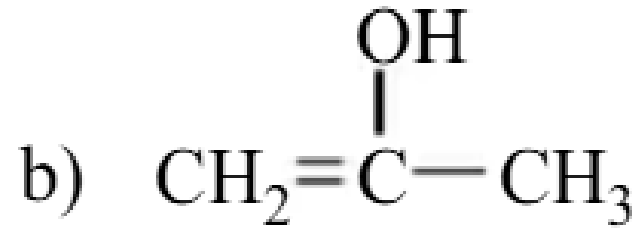
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EXERCISE

What is the product when  is reduced using H_2 gas and a Ni catalyst?

a) 

c) 

b) 

d) 

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EXERCISE

Which of the following compounds would be resistant to oxidation with a mild oxidizing agent?

a) $CH_3-CH_2-CH_2-OH$

b) $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-H$

c) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

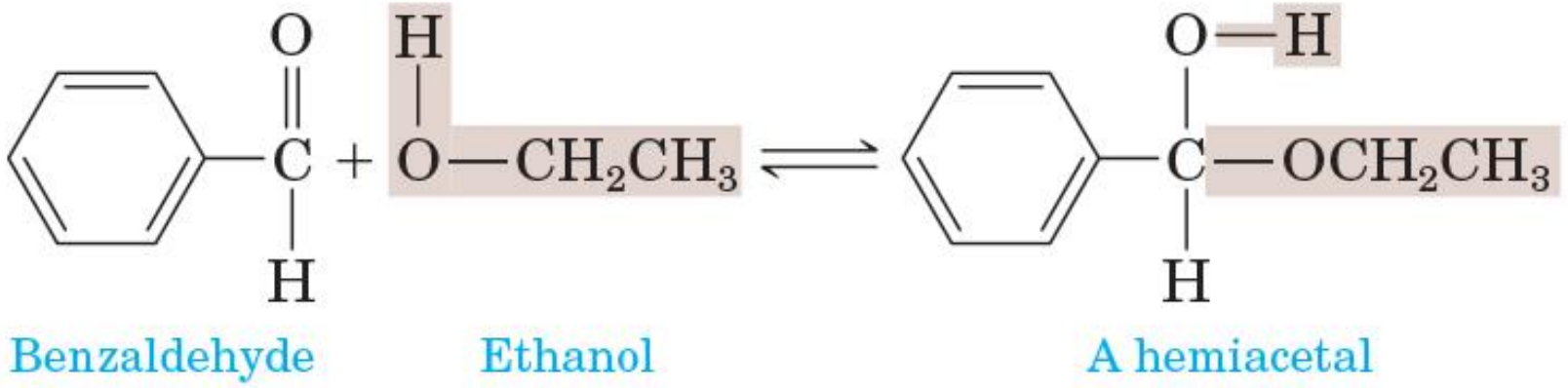
d) more than one correct response

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Addition of Alcohols

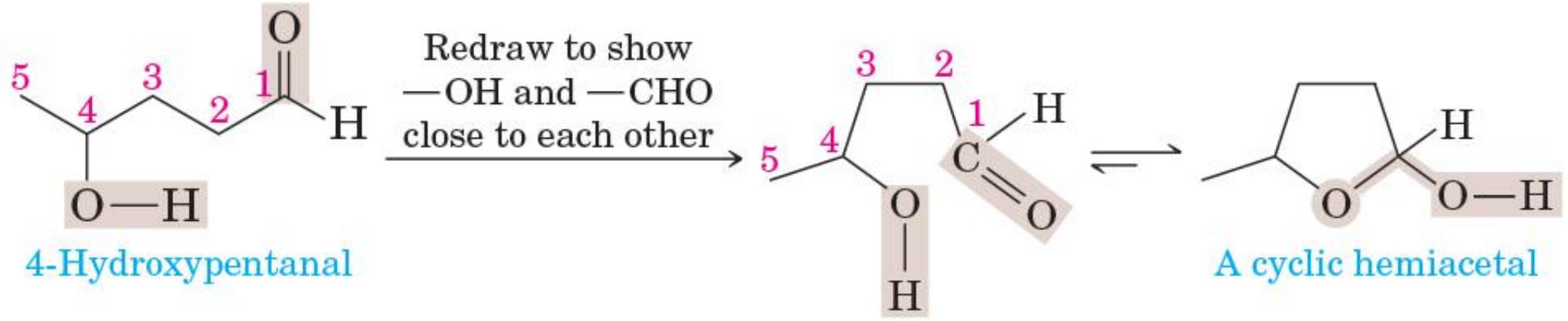
Addition of a molecule of alcohol to the carbonyl group of an aldehyde or ketone forms a **hemiacetal** (a half-acetal).

- The functional group of a hemiacetal is a carbon bonded to one –OH group and one -–OR group.
- In forming a hemiacetal, –H of the alcohol adds to the carbonyl oxygen and –OR adds to the carbonyl carbon.



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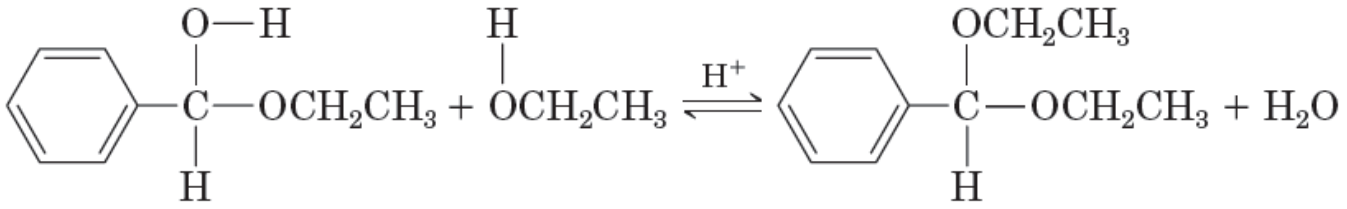
- Hemiacetals are generally unstable and are only minor components of an equilibrium mixture except in one very important type of molecule.
- When a hydroxyl group is part of the same molecule that contains the carbonyl group and a five- or six-membered ring can form, the compound exists almost entirely in a cyclic hemiacetal form.

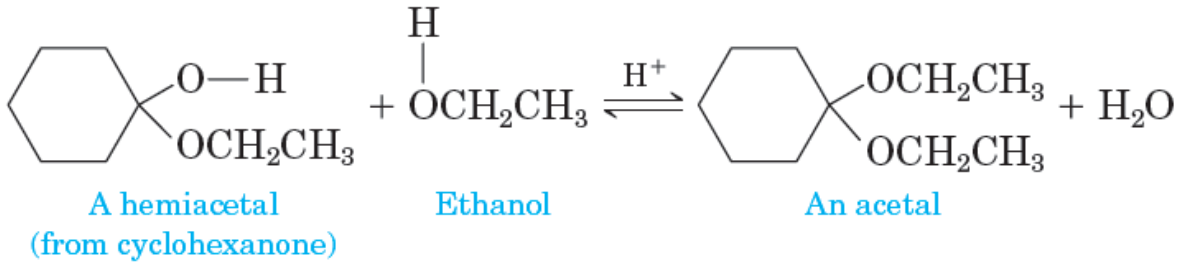


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A hemiacetal can react further with an alcohol to form an **acetal** plus water.

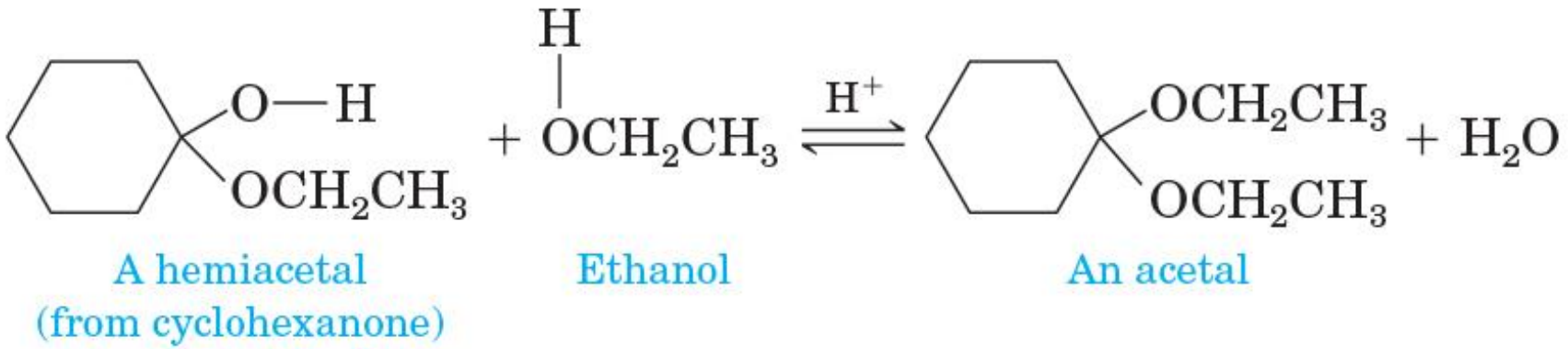
- This reaction is acid-catalyzed.
- The functional group of an acetal is a carbon bonded to two –OR groups.





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- All steps in hemiacetal and acetal formation are reversible.
- As with any other equilibrium, we can drive it in either direction by using Le Chatelier's principle.
- To drive it to the right, we either use a large excess of alcohol or remove water from the equilibrium mixture
- To drive it to the left, we use a large excess of water.

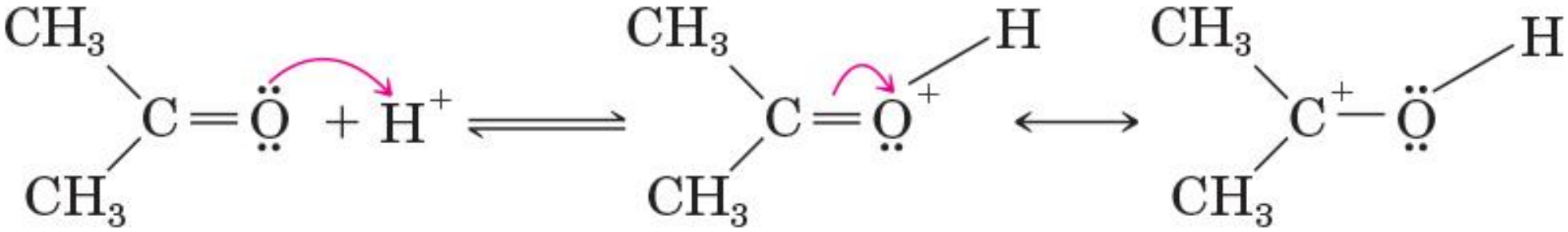


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Mechanism of Acetal Formation

Step 1: Add a proton.

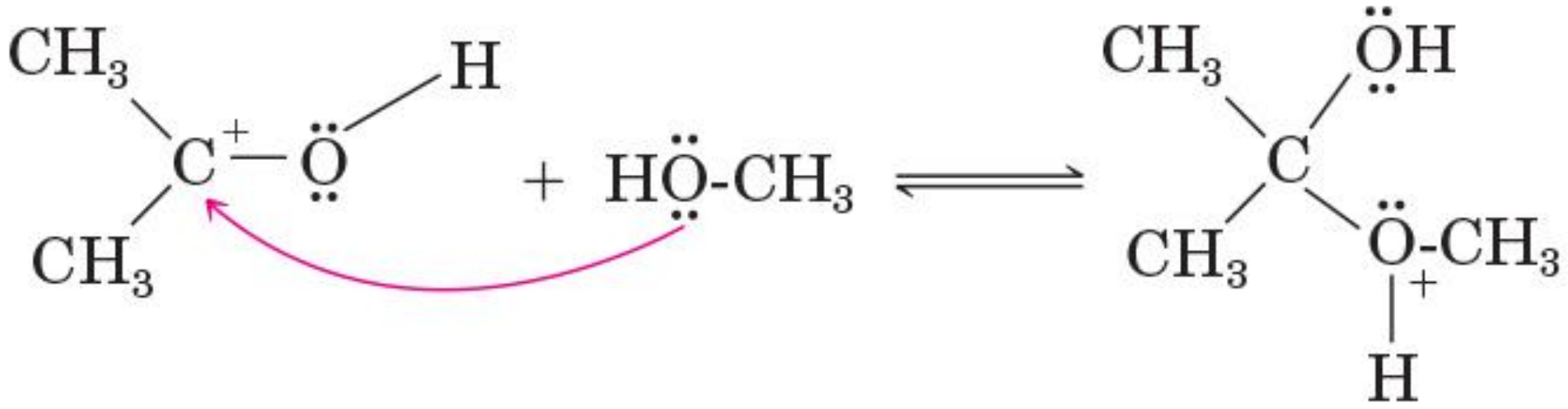
Adding a proton to the carbonyl oxygen makes the carbonyl carbon stronger electrophile and more susceptible to attack by a nucleophile.



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Step 2: Reaction of an electrophile and a nucleophile to form a new covalent bond.

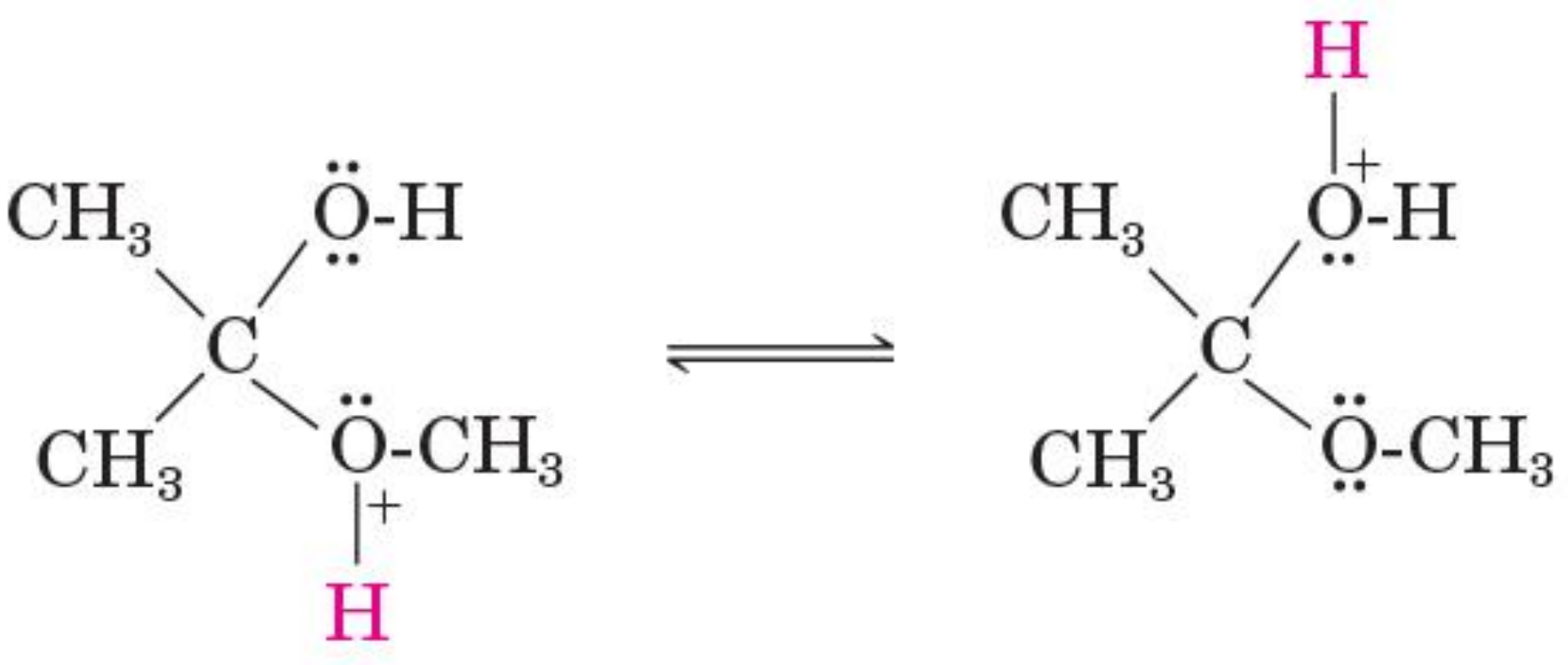
This step adds the first –OR group required for acetal formation. The intermediate formed is an oxonium ion.



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Step 3: Proton transfer to another oxygen.

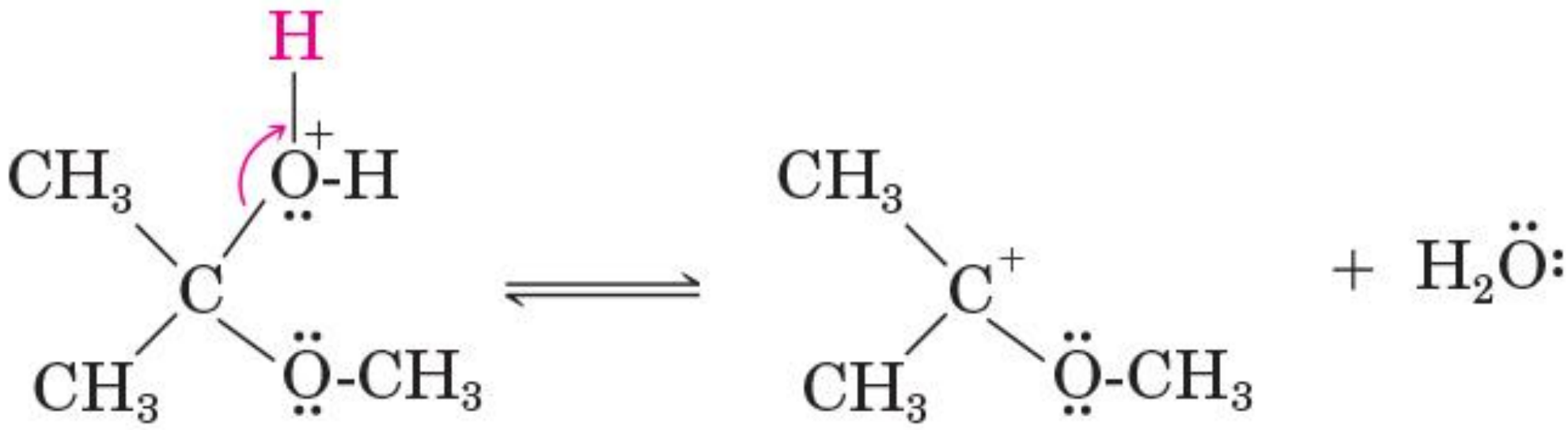
The proton transferred is in red.



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Step 4: Break a bond to form stable molecules or ions.

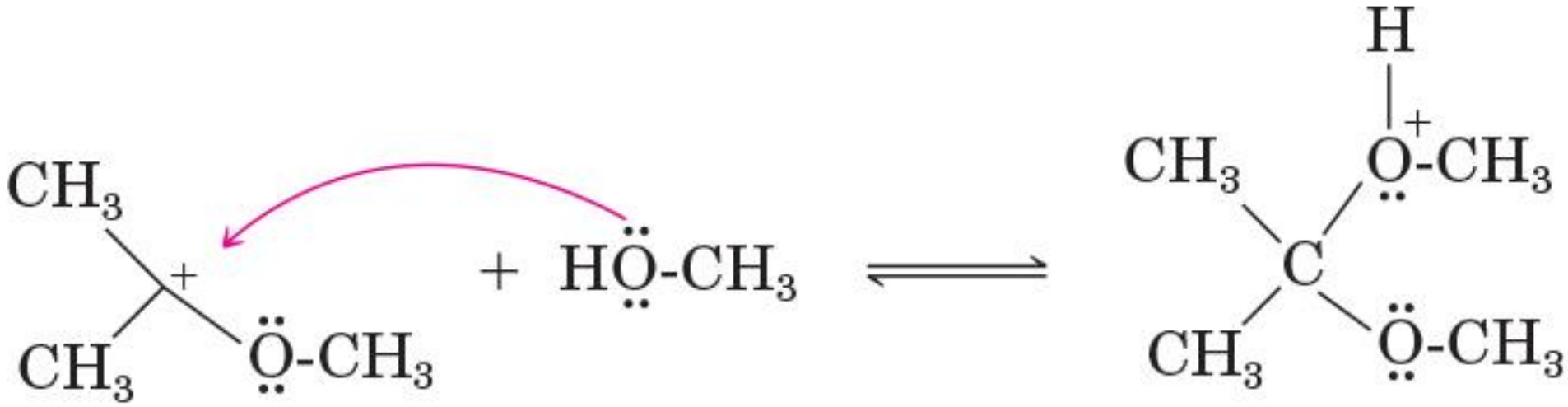
In this case the stable molecule is H_2O and the stable ion is a 3°carbocation.



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Step 5: Reaction of an electrophile and a nucleophile to form a new covalent bond.

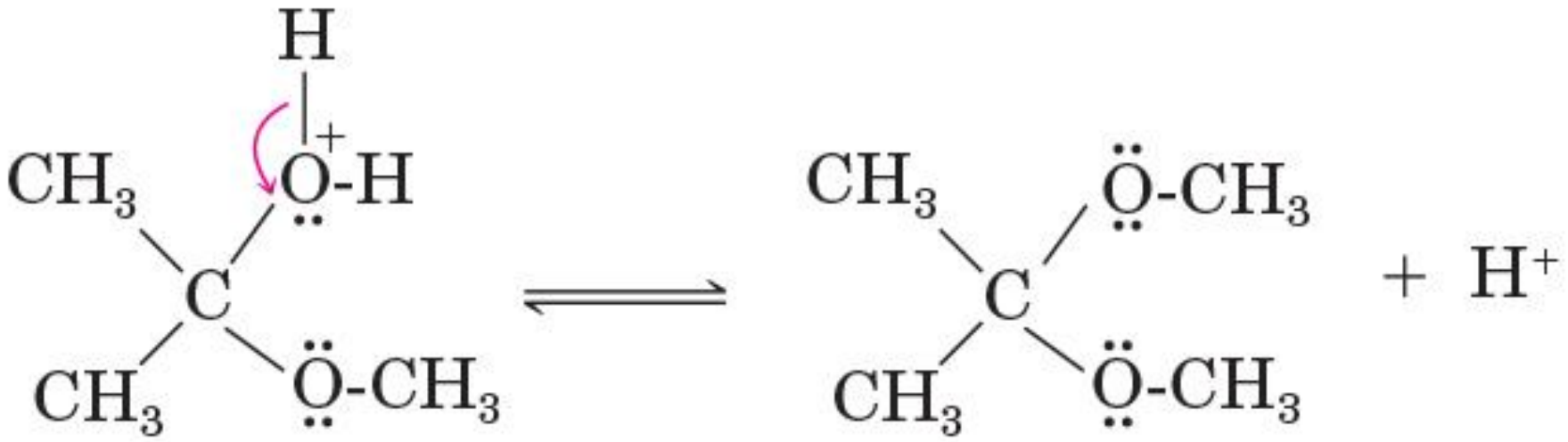
This step adds the second –OR group of the acetal to what was the original carbonyl carbon.



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Step 6: Take a proton away.

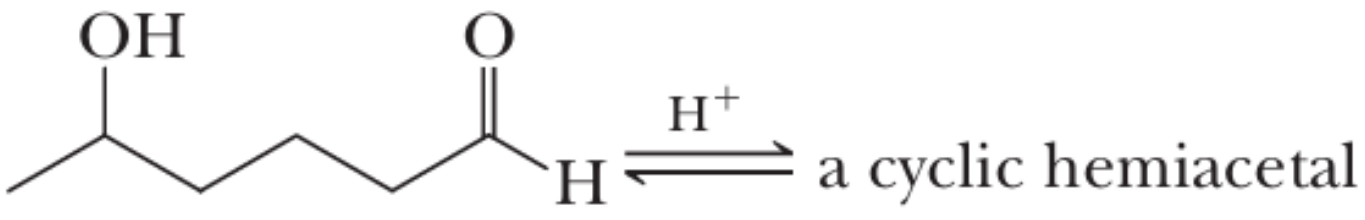
Transfer of a proton to solvent gives the acetal and regenerates the H^+ catalyst.



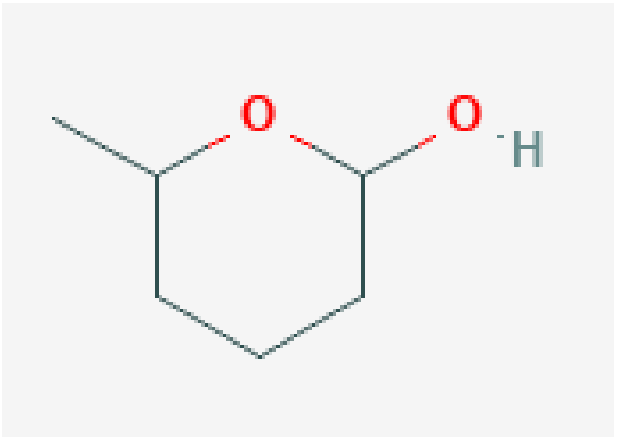
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EXERCISE

5-Hydroxyhexanal forms a six-membered cyclic hemiacetal, which predominates at equilibrium in aqueous solution. Draw a structural formula for this cyclic hemiacetal.



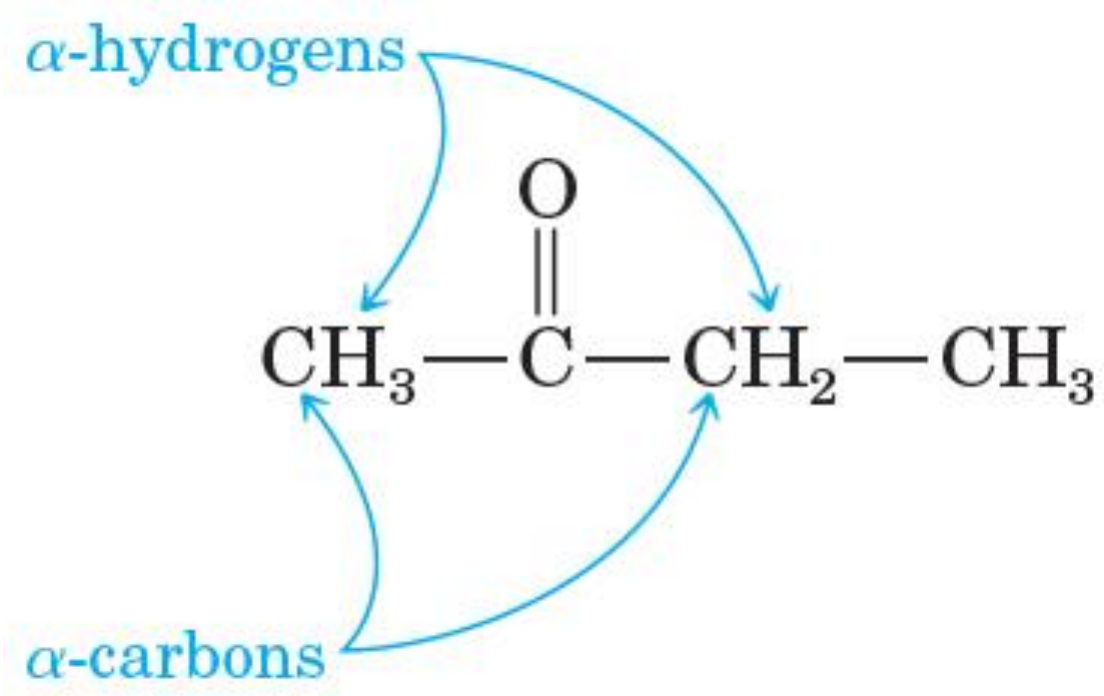
5-Hydroxyhexanal



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Keto-Enol Tautomerism

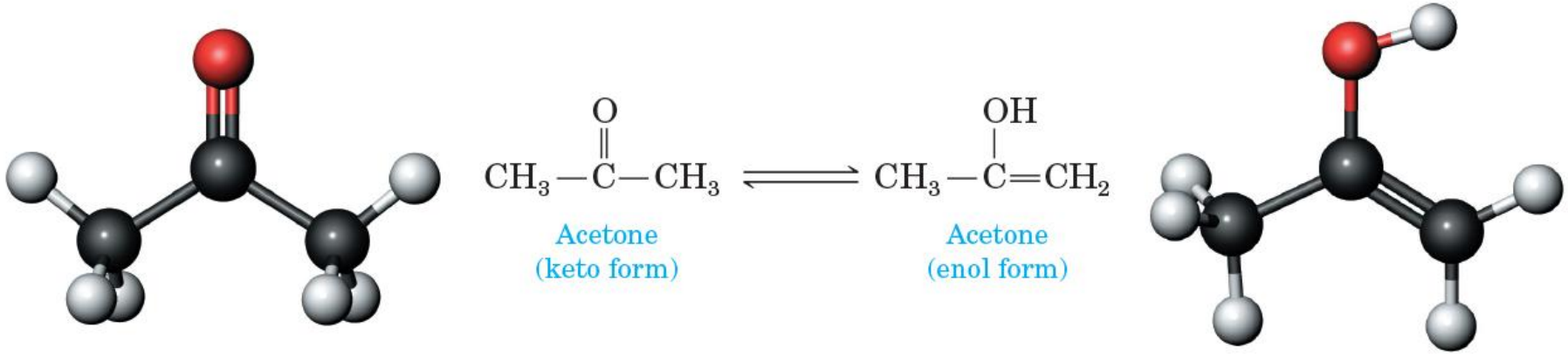
A carbon atom adjacent to a carbonyl group is called an **α-carbon**, and a hydrogen atom bonded to it is called an **α-hydrogen**.



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An aldehyde or ketone that has a hydrogen on an α-carbon is in equilibrium with a constitutional isomer called an **enol**.

- The name “enol” is derived from the IUPAC designation of it as both an alkene (-**en**-) and an alcohol (-**ol**).



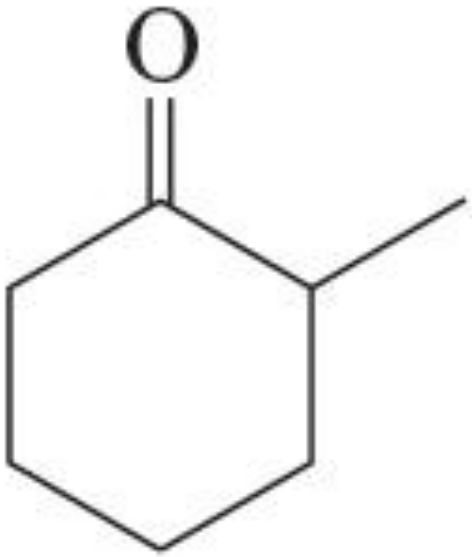
  - In a keto-enol equilibrium, the keto form generally redominates.

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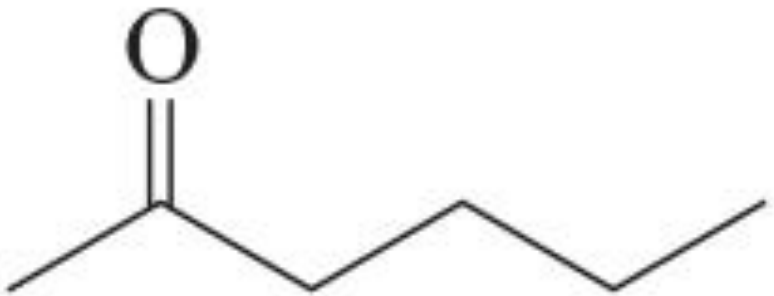
EXERCISE

Draw structural formulas for the two enol forms for each ketone.

(a)



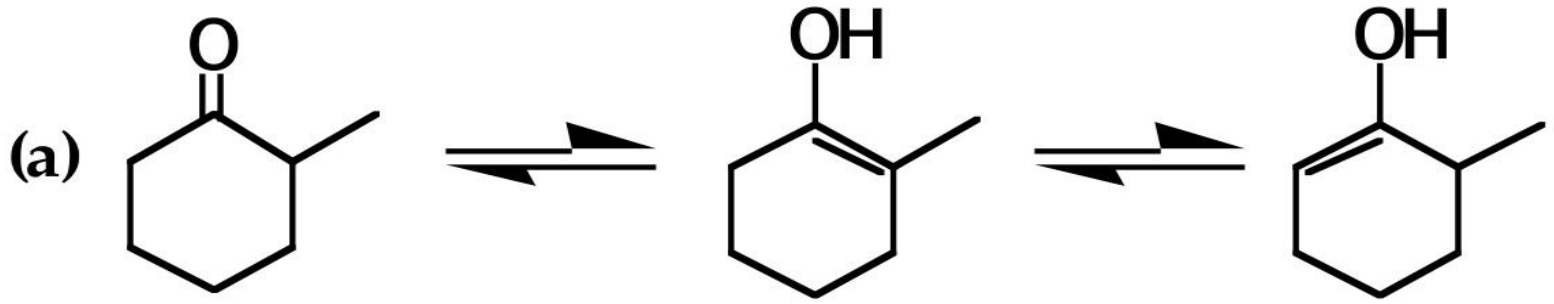
(b)

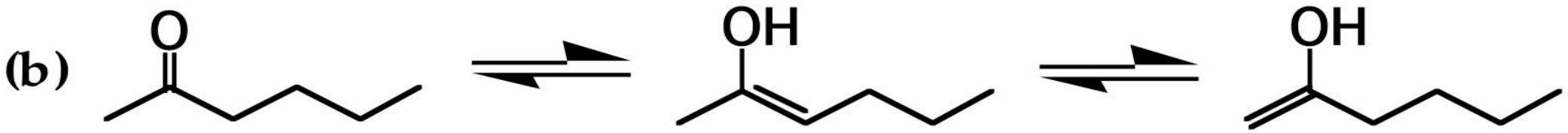


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ANSWER

Solution:





cis and *trans* isomers are possible for this enol

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